THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The radii of two solid spheres are in the ratio 2 : 3. The ratio of their volumes is: 1

(a) 8 : 27 **(b)** 16 : 27 **(c)** 2 : 3 **(d)** 4 : 9

2. For a distribution with median class 20–30, $N = 50$, $cf = 18$ and $f = 14$, the median is: 1

(a) 28 **(b)** 25 **(c)** 24 **(d)** 22.5

3. A box contains 70 tickets numbered 1 to 70. One ticket is drawn at random. The probability that the number drawn is a prime number is: 1

(a) $\frac{9}{35}$ **(b)** $\frac{2}{7}$ **(c)** $\frac{51}{70}$ **(d)** $\frac{19}{70}$

4. A solid cylinder of base radius 6 cm and height 10 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 4 cm. The number of cones obtained is: 1

(a) 60 **(b)** 34 **(c)** 10 **(d)** 30

5. The mean of a distribution with classes 100–120, 120–140, 140–160, 160–180, 180–200, 200–220, 220–240, 240–260 and frequencies 19, 3, 4, 12, p, 13, 22, 7 respectively is 184.6. The value of p is: 1

(a) 22 **(b)** 19 **(c)** 20 **(d)** 25

6. A lot of 250 ball pens contains 35 defective pens. One pen is drawn at random from the lot. The probability that the pen is good (not defective) is: 1

(a) $\frac{43}{50}$ **(b)** $\frac{107}{125}$ **(c)** $\frac{108}{125}$ **(d)** $\frac{7}{50}$

7. A hemispherical bowl has outer radius 9 cm and inner radius 7.5 cm. The volume of material used to make the bowl is: 1

(a) 321.75 cm³ **(b)** 643.5 cm³ **(c)** 1287 cm³ **(d)** 1930.5 cm³

8. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 20, x, 18, y, 12, 7, 18 respectively, and total frequency 100, is 25. The value of x is: 1

(a) 20 **(b)** 22 **(c)** 21 **(d)** 6

9. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the sum of the numbers is 7 is: 1

(a) $\frac{7}{36}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{5}{6}$

10. The largest possible solid sphere is carved out of a cube of edge 42 cm. The volume of the material wasted is: 1

(a) 38808 cm³ **(b)** 74088 cm³ **(c)** 35280 cm³ **(d)** 54684 cm³

11. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 with frequencies 21, 13, 25, 5, 6, 24, 4, 2 respectively, The mean of the data is: 1

(a) 85 **(b)** 71 **(c)** 72 **(d)** 887.5

12. Two fair dice are thrown together and the numbers on the upper faces are noted. The probability that the numbers differ by 2 is: 1

(a) $\frac{7}{36}$ **(b)** $\frac{7}{9}$ **(c)** $\frac{2}{9}$ **(d)** $\frac{1}{4}$

13. **Assertion (A):** A cone and a cylinder have the same base radius and the same height. The volume of the cone is one-third that of the cylinder. 1

Reason (R): The volume of a cone of base radius r and height h is $\pi r^2 h$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** For grouped data the median class is the class whose cumulative frequency is just greater than or equal to $\frac{n}{2}$. 1

Reason (R): The median class is the class which has the highest frequency.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** The probability of getting a prime number when a die is thrown once is $\frac{1}{2}$. 1

Reason (R): The probability of an event can be greater than 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A bag contains 18 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. Find the value of x. 2

17. A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. Find the length of the wire. 2

18. The marks obtained by students in a test are given below. 2

Marks	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of students	23	4	25	6	3	19	20

Find the median.

19. A bag contains 15 red and 9 blue balls. 6 more red balls are put into the bag and a ball is drawn at random. Find the probability that the ball drawn is red. 2

20. A solid sphere of radius 9 cm is completely dropped into water in a cylindrical vessel of base radius 12 cm, and no water spills. Find the rise in the water level. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 17, 7, x, 20, y, 19, 17 respectively, and total frequency 100, is 239.5. Find the value of x. 3

22. A box contains 84 tickets numbered 1 to 84. One ticket is drawn at random. Find the probability that the number drawn is an even prime number. 3

23. A solid metallic cylinder of base radius 6 cm and height 72 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

24. The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	20–40	40–60	60–80	80–100	100–120	120–140	140–160
Number of workers	3	13	11	17	15	24	17

Find the mean and the mode (in this order).

OR

Two fair dice are thrown together and the numbers on the upper faces are noted. Find the probability that both numbers are even. 3

25. A solid metallic cone of base radius 8 cm and height 4 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

OR

The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of workers	10	22	24	6	21	2	15

Find the mean.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A teacher recorded the time (in minutes) taken by 40 students to solve a puzzle. The classes 10-20, 20-30, 30-40, 40-50 and 50-60 had frequencies 2, 4, 10, 14 and 10 respectively. 5

(i) Find the class mark of the class 30-40.

(ii) Write the modal class.

(iii) Find the mean time taken.

(iv) Find the mode of the data.

27. A toy is a cone of base radius 63 cm and height 84 cm mounted on a hemisphere of the same radius (the flat faces coincide). Find the total surface area of the toy. 5

OR

The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 20, x, 18, y, 12, 7, 18 respectively, and total frequency 100, is 25. Find the value of x. 5

END OF QUESTION PAPER — Total: 50 Marks